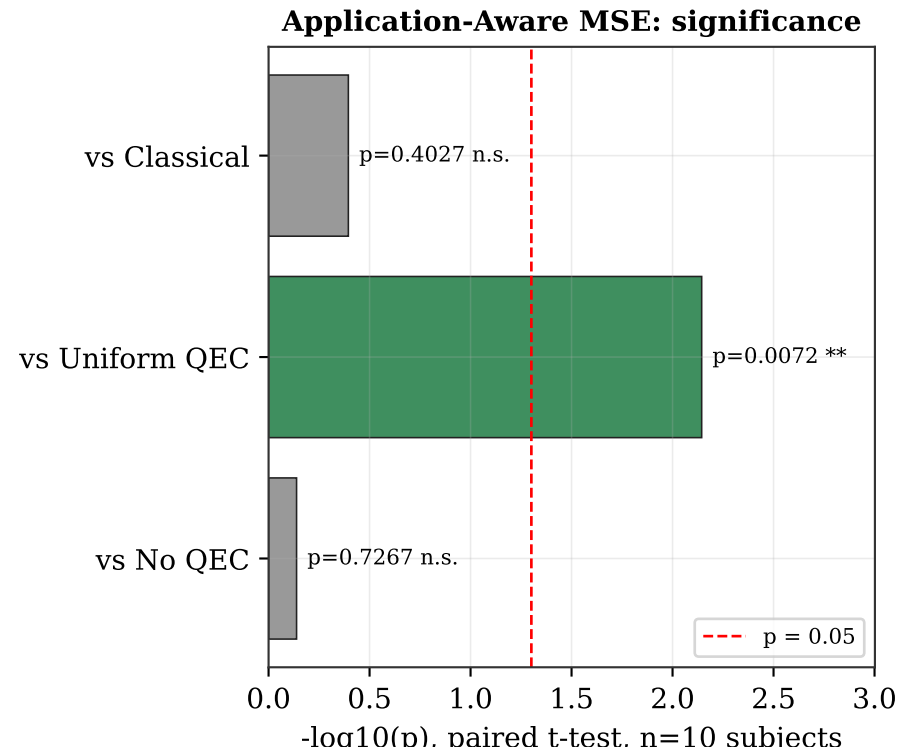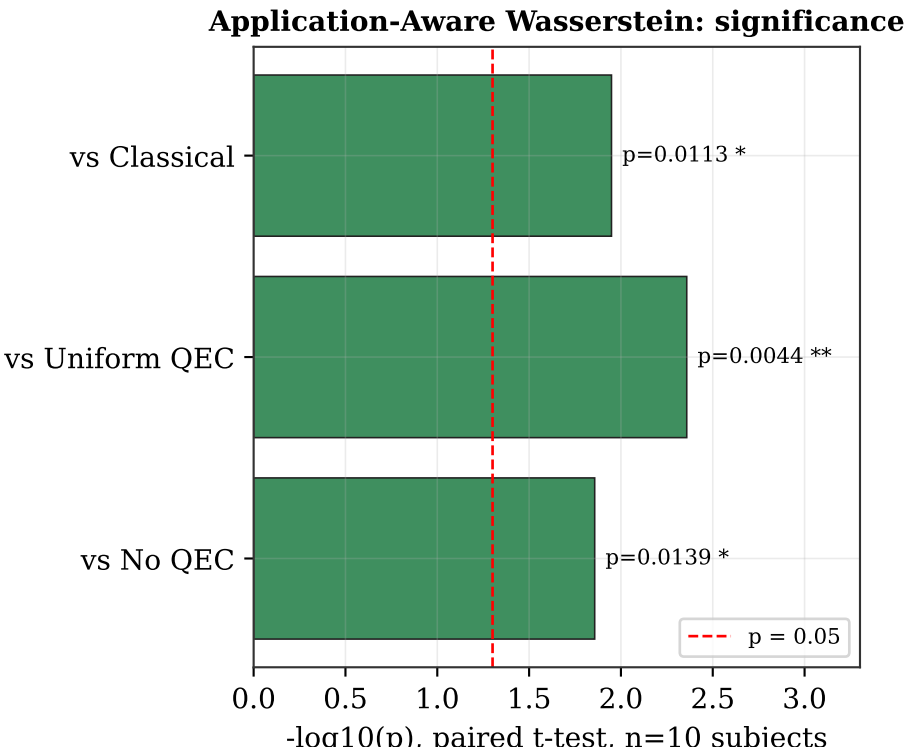Statistical Significance of Application-Aware QEC's Advantage